THE TOP LOCALIZED CHERN CLASS OF THE SECTION OF A PROJECTIVE BUNDLE VANISHES BELOW THE RANK AND IS A FACTORIAL MULTIPLE OF THE VECTOR BUNDLE PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FEF (sources/stacks-chow/pinned-original-source.tex line 12542).

Allowed standard prerequisites (editorial).

- The Koszul resolution of the section of a projective bundle and the computation of its Chern classes. Same chapter, Lemma labelled lemma-compute-koszul, cited at proof lines 12577-12580. Shows i_* O_X is perfect, gives the vanishing below r and computes c_r as a factorial multiple.
- Chern classes of a dual and of a bundle with a regular section. Same chapter, Lemma labelled lemma-chern-classes-dual, cited at proof lines 12582-12584, and Lemma labelled lemma-top-chern-class, cited at proof lines 12604-12606. Rewrites the class into the form whose zero locus is i(X) and evaluates it on integral subschemes.
- Cycle classes on a vector bundle and the restriction sequence to its complement. Same chapter, Lemma labelled lemma-vectorbundle, cited at proof lines 12592-12594, and Lemma labelled lemma-restrict-to-open, cited at proof lines 12594-12596. Decomposes alpha as pi^* beta + gamma with gamma supported off C.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FEE: sources/stacks-chow/pinned-original-source.tex, source lines 12500–12500. Same-chapter pointer into the author environment lemma

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at source line 12500; invoked at proof line 12574 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FA5: sources/stacks-chow/pinned-original-source.tex, source lines 8520–8520. Same-chapter pointer into the author environment lemma at source line 8520; invoked at proof line 12580 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02TY: sources/stacks-chow/pinned-original-source.tex, source lines 7062–7062. Same-chapter pointer into the author environment lemma at source line 7062; invoked at proof line 12593 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02RX: sources/stacks-chow/pinned-original-source.tex, source lines 2949–2949. Same-chapter pointer into the author environment lemma at source line 2949; invoked at proof line 12594 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FA9: sources/stacks-chow/pinned-original-source.tex, source lines 8750–8750. Same-chapter pointer into the author environment lemma at source line 8750; invoked at proof line 12605 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $\mathcal{C}$ be a locally free $\mathcal{O}_X$-module of rank r. Consider the morphisms*

$$X = \underline{Proj}_X(\mathcal{O}_X[T]) \xrightarrow{i} E = \underline{Proj}_X(Sym^*(\mathcal{C})[T]) \xrightarrow{\pi} X$$

Then $c_t(i_*\mathcal{O}_X) = 0$ for $t = 1, \ldots, r-1$ and in $A^0(C \to E)$ we have

$$p^* \circ \pi_* \circ c_r(i_*\mathcal{O}_X) = (-1)^{r-1}(r-1)!j^*$$

where $j : C \to E$ and $p : C \to X$ are the inclusion and structure morphism of the vector bundle $C = \underline{\mathrm{Spec}}(Sym^*(\mathcal{C}))$.

Proof. The canonical map $\pi^*\mathcal{C} \to \mathcal{O}_E(1)$ vanishes exactly along $i(X)$. Hence the Koszul complex on the map

$$\pi^*\mathcal{C} \otimes \mathcal{O}_E(-1) \to \mathcal{O}_E$$

is a resolution of $i_*\mathcal{O}_X$. In particular we see that $i_*\mathcal{O}_X$ is a perfect object of $D(\mathcal{O}_E)$ whose Chern classes are defined. The vanishing of $c_t(i_*\mathcal{O}_X)$ for $t = 1, \ldots, t-1$ follows from Lemma 0FEE. This lemma also gives

$$c_r(i_*\mathcal{O}_X) = -(r-1)!c_r(\pi^*\mathcal{C} \otimes \mathcal{O}_E(-1))$$

On the other hand, by Lemma 0FA5 we have

$$c_r(\pi^*\mathcal{C} \otimes \mathcal{O}_E(-1)) = (-1)^r c_r(\pi^*\mathcal{C}^\vee \otimes \mathcal{O}_E(1))$$

and $\pi^*\mathcal{C}^\vee \otimes \mathcal{O}_E(1)$ has a section s vanishing exactly along $i(X)$.

After replacing X by a scheme locally of finite type over X, it suffices to prove that both sides of the equality have the same effect on an element $\alpha \in \mathrm{CH}_*(E)$. Since $C \to X$ is a vector bundle, every cycle class on C is of the form $p^*\beta$ for some $\beta \in \mathrm{CH}_*(X)$ (Lemma 02TY). Hence by Lemma 02RX we can write $\alpha = \pi^*\beta + \gamma$ where γ is supported on $E \setminus C$. Using the equalities above it suffices to show that

$$p^*(\pi_*(c_r(\pi^*\mathcal{C}^\vee \otimes \mathcal{O}_E(1)) \cap [W])) = j^*[W]$$

when $W \subset E$ is an integral closed subscheme which is either (a) disjoint from C or (b) is of the form $W = \pi^{-1}Y$ for some integral closed subscheme $Y \subset X$. Using the section s and Lemma 0FA9 we find in case (a) $c_r(\pi^*\mathcal{C}^\vee \otimes \mathcal{O}_E(1)) \cap [W] = 0$ and

in case (b) $c_r(\pi^*\mathcal{C}^\vee \otimes \mathcal{O}_E(1)) \cap [W] = [i(Y)]$. The result follows easily from this; details omitted. $\square$